A1 Reproduction and Verification

Commands, environment and what the automatic checks do and do not cover

Step	What	Command (PowerShell, repository root)
1 Environment	Python 3.12.3, CPU; exact run versions in requirements-lock.txt and run.json	`pip install -r requirements-lock.txt -r requirements-timestral.txt`
2 Data (not in Git)	NQ 1-minute bars 2006–2025 (TradeStation, licensed); Timestral-5M weights in the local Hugging Face cache	`$env:FUTURES_DATA_DIR = "C:\path\to\futures-data"`
3 Train and forecast	all models, five timeframes	`python multitarget.py --tag final`
4 Tables and charts	results, rankings, timeframe map, charts, deck_data.json	`python multitarget_report.py <run> reports\multitarget_final`

5 Sample and Timestral checks	appendix A3, A4, A6 (1-minute data, saved forecasts, Yahoo daily data)	`python tools\sample_checks.py <run> reports\multitarget_final`
		`python tools\timestral_same_sample.py <run> reports\multitarget_final`
		`python tools\yahoo_compare.py data_raw\yahoo reports\multitarget_final presentation`

6 Copy, verify, build	copy deck_data.json, charts and the three JSON files to presentation\	`python verify_multitarget.py <run> reports\multitarget_final presentation` `cd presentation; npm ci; node build_deck.js`

Verified

- every point estimate in the results tables, recomputed independently from the saved forecasts
- the ranking rules, the five-model tables, the timeframe map, the data profile and the deck data
- past-to-future Timestral calibration: no calibration target after the forecasts it is used for
- sample-check content (A3): flags vs threshold and their predecessor, categories, daily accounting vs saved forecasts
- unit tests (pytest) for targets, timing, splits, bootstrap, calibration and flag provenance

Not verified

- the raw 1-minute data and the candle building (measured separately in A3, not re-read by the verifier)
- the training: models were not re-fitted
- bootstrap intervals: only their order is checked
- the run used an uncommitted version of multitarget.py (hash in run.json); later changes: majority-class fix applied to saved outputs, docstrings

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A2 Selection and Statistics

How models were tuned, ranked and given uncertainty intervals

- **Tuning:** hyper-parameters (Ridge α, logistic C) and early stopping (XGBoost, networks) use the validation block (2017-08-25 → 2021-07-14) only.
- **Evidence score z:** paired validation improvement over the task's naive benchmark (squared-error reduction vs zero; accuracy gain vs majority; squared log-RV error reduction vs persistence) divided by its moving-block bootstrap standard error (20-session blocks, 2,000 replicates).
- **Revision (not blind):** the first volatility score was the QLIKE reduction (Patton, 2011). On validation a few limit-locked March 2020 candles with almost zero variance dominated it, so after a first report had printed test numbers it was replaced by the log-RV squared-error reduction.
- **Five-model tables (slides 6–8):** one model per family on each task's best validation timeframe, ranked by z. Timeframe map (slide 4): the highest-z trained model per task and timeframe.
- **Intervals:** differences (accuracy gains) use the bootstrap of paired per-row differences; ratios (R^2-type error cuts) resample model and benchmark losses together and recompute the ratio in each replicate (Künsch, 1989).
- **Majority-class fix:** the first run counted dojis as "not up"; the majority class was recomputed from non-doji training candles without refitting any model (tools/mt_fix_majority.py).

A3 Sample Checks

Why the daily test ends on 2025-01-30, and whether incomplete candles matter

A. Daily sample end (measured, tools/sample_checks.py)

- **Cause:** all 85 sessions from 2025-01-31 to 2025-05-30 are dropped; 84 fall within 50 sessions after a session with a flagged 1-minute bar (|log return vs the previous bar| > 2%).
- **What generated each flag:** the flag is computed before out-of-schedule bars are dropped. Measured on that sequence, all 21 flags (16 in schedule) exceed the threshold.
- **Verified facts:** 5 in-schedule flags (e.g. 2023-03-16, +7.2%) are measured against a dropped 16:59 CT bar 989–991 points below the bars just before and after it; against the last kept bar they move < 0.05%. 9 move more than 2% and are still beyond 2% 30 minutes later (5 of them after a weekend gap). 2 partly retrace (2020-03-02, 2020-03-16): unresolved.
- **Clues, not proof:** several persistent moves coincide with the 8:30 ET data releases (2022) or a weekend reopening; the five 16:59 CT bars look like data artefacts. Neither timing nor persistence proves that a price is genuine or wrong.
- **Impact:** 304 of 1,002 daily test sessions lie in such 50-session windows and are excluded (50 of them only because of the 2023-03-16 flag of the first kind). Another 56 sessions are excluded because the next session is not the label (conditions overlap); 642 are kept. The daily sample under-represents turbulent periods.
- **Not corrected:** the reconciled daily returns of the excluded sessions agree with Yahoo (A6): the quarantine also removes plausible sessions (minutes and RV not verified). Changing the rule needs a new daily run (not done); the saved results are kept, exploratory, for the filtered sample.

B. Full-length candles with missing 1-minute bars

Candle	All years	Test	Test, 8:30–15:00 CT	Test labels affected	DNN 128 error cut (%) all → complete
5 min	18.0%	0.23%	0.003%	601	39.99 → 40.00
15 min	25.8%	0.43%	0.008%	385	35.06 → 35.07
30 min	30.6%	0.65%	0.008%	279	41.72 → 41.80
1 hour	34.3%	1.01%	0.000%	201	41.97 → 42.24

- A full-length candle with 4 of its 5 minutes is accepted as complete; the count is stored (n_minutes) but not used (unit test documents it).
- Missing minutes are mostly overnight in 2006–2017, consistent with thin trading (no bar is written for a minute without trades), and almost absent in regular hours. In the test, a few sessions with long gaps coincide with quarterly contract-roll days (e.g. 2021-09-09, 2022-09-08), consistent with dataset gaps.
- Scope: the SAVED test forecasts rescored without rows whose label candle is incomplete; the rescored metrics (HAR-OLS, DNN, XGBoost) move by at most 0.4 points. Conclusions unchanged in this test-sample check; the effect on training and on the features of kept rows, and a refit on corrected data, were not evaluated.

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A4 Timestral on the Same Sample

Mistral's Timestral-5M vs the trained models on exactly the same random test candles (saved forecasts)

Candle	Sample	Return R^2: Timestral	Return R^2: DNN	Direction pp: Timestral, body	Direction pp: Timestral, close-to-close	Direction pp: DNN, body	Vol. error cut: Timestral	Vol. error cut: DNN	Corr. with next log RV: spread / current RV
5 min	20,000	−2.55%	+0.20%	−0.59	−0.66	+0.51	+12.2%	+39.6%	0.76 / 0.76
15 min	20,000	−2.23%	−0.10%	+0.01	−0.08	+1.09	−37.1%	+34.8%	0.74 / 0.84
30 min	20,000	−1.46%	+0.02%	−1.76	−1.88	+0.14	−56.2%	+41.3%	0.71 / 0.84
1 hour	20,000	−1.59%	+0.00%	−2.28	−2.36	+0.02	−74.3%	+41.8%	0.64 / 0.83
1 day	642	−1.83%	+0.17%	−1.87	−1.56	+0.00	−19.3%	+18.7%	0.75 / 0.82

- **Target mismatch is not the reason:** Timestral forecasts the next close, the study scores the candle body; scored on the close-to-close direction it does no better (columns 5–6).
- **Same rows:** the trained models keep their advantage on Timestral's own sample, so the gap is not a sampling artefact.
- **Volatility is indirect:** Timestral's volatility comes from the spread of its PRICE forecast (q10–q90), mapped to log RV by a linear calibration; the trained models forecast log RV directly from past RV. The raw spread correlates with next log RV slightly less than current RV does, and the calibration fails beyond 5 minutes.
- **Scope:** one small model (5M), frozen weights, 512-close context, one market: not a verdict on foundation models in general.
- **Proposed next experiment (not run):** aligned targets — forecast open and close of t+1 (body direction) and the quantiles of the next candle's 1-minute returns, then derive RV from Timestral's own path; compare 28M–209M on the same sample.

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A5 Data Profile

Descriptive statistics of the three targets, validation and test blocks (data_profile.csv)

Block	Candle	Rows	Sessions	UP / DOWN / doji	Return s.d. (bp)	ACF return	Mean log RV	RV rises	ACF log RV
Validation	5 min	269,167	2017-08-25 → 2021-07-14	48.8% / 47.2% / 4.0%	7.0	−0.006	-16.13	48.9%	+0.76
Validation	15 min	88,336	2017-08-25 → 2021-07-14	50.5% / 47.3% / 2.2%	12.0	+0.003	-14.81	47.9%	+0.85
Validation	30 min	44,513	2017-08-25 → 2021-07-14	51.3% / 47.1% / 1.6%	16.3	−0.001	-14.06	46.2%	+0.85
Validation	1 hour	21,642	2017-08-25 → 2021-07-14	52.5% / 46.3% / 1.2%	22.7	+0.011	-13.28	47.3%	+0.84
Validation	1 day	822	2017-08-25 → 2021-07-13	57.7% / 41.8% / 0.5%	101.5	−0.052	-9.69	45.4%	+0.84
Test	5 min	273,340	2021-07-15 → 2025-05-30	49.9% / 48.3% / 1.8%	7.7	−0.005	-15.75	48.9%	+0.75
Test	15 min	90,189	2021-07-15 → 2025-05-30	50.7% / 48.3% / 1.0%	13.3	+0.003	-14.45	47.3%	+0.84
Test	30 min	44,395	2021-07-15 → 2025-05-30	51.1% / 48.1% / 0.8%	18.5	−0.002	-13.69	46.0%	+0.84
Test	1 hour	21,507	2021-07-15 → 2025-05-30	51.5% / 48.0% / 0.5%	26.2	−0.000	-12.90	47.7%	+0.83
Test	1 day	642	2021-07-15 → 2025-01-30	54.2% / 45.8% / 0.0%	116.9	−0.001	-9.26	49.1%	+0.81

ACF = lag-1 autocorrelation (intraday: pairs within a session; daily: consecutive sessions). RV = sum of squared 1-minute log returns inside the candle.

A6 External Comparison: Yahoo Finance

Daily candles vs ^NDX (cash index) and NQ=F (front-month futures, not adjusted): tools/yahoo_compare.py

A. Daily returns of the excluded sessions vs Yahoo

Daily sessions	With a quote	Returns agree	Median gap to ^NDX
Test, in 50-session windows (excluded)	292 of 304	100.0%	0.034 pp
Test, outside the windows	658 of 698	99.5%	0.030 pp
All splits, in the windows	551 of 569	99.6%	0.032 pp
All splits, outside	4,214 of 4,442	99.7%	0.032 pp

- **Rule:** "agree" = re-levelled close-to-close return within 0.5 pp of ^NDX (15:00 CT close) or of NQ=F (session close). Closing returns only: the range is not in the rule, so the candle, single minutes and RV are not validated.
- **Flagged sessions:** returns agree for 16 of 16 (2023-03-16: +2.67% vs ^NDX +2.66%; 2020-03-16: −13.43% vs ^NDX −13.00%); their ranges are also close to NQ=F (descriptive).
- **Disagreeing:** 13 sessions in 2006–2025 (test: 2023-09-15, 2023-12-18, 2024-12-23), mostly quarterly-expiry days; none is excluded.
- **Conclusion:** the reconciled daily returns of the excluded sessions agree with the Yahoo references within the chosen tolerance. This suggests that the quarantine also removes plausible sessions, but it does not verify the single minutes or the realized volatility. Not a data-quality certification; no filter changed.

B. Additive back-adjustment compresses % returns

Period	S.d. ratio vs ^NDX (as used)	S.d. ratio, re-levelled	Correlation, re-levelled
2006–2012	0.42	0.98	0.996
2013–2020	0.74	1.00	0.999
Train + validation	0.59	0.99	0.998
Test	0.90	0.99	0.998

- TradeStation adds every roll gap to earlier prices: the level is about 2,206 points above the unadjusted price in 2010, 771 in 2024. Point changes are kept, percentages are not.
- Re-levelled = a retrospective estimate (offset: centred 21-day median of Yahoo NQ=F minus futures-data), not the contracts' original prices nor an independent validation. After it, daily returns are close to ^NDX (correlation 0.998, beta 0.99).
- Return and log-RV levels and the training history are scaled (log-RV shift ≈ −1.7 in 2006–2012, −0.3 to 0 in the test); the factor changes over time and between training and test. The effect on relative performance was not measured; a fix needs ratio adjustment and a new run.